

ETHAN C. GOLDSTEIN

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Computer Information Systems student who builds systems and the validation that checks them. Built Autonomous OS, an 11-agent orchestration platform of roughly 12,800 lines running unattended, with a human approval gate before every outbound action.

EDUCATION

University of South Carolina

B.S. Computer Information Systems, Minor in Business Information Management

Coursework: Data Structures and Algorithms (Java), Information Security Principles, Web Applications, Capstone Project

Study Abroad: Florence University of the Arts, Firenze, Italy

Columbia, SC

Expected May 2027

Spring 2026

EXPERIENCE

AI & Workflow Automation Engineer

2025 to Present

Independent Practice · github.com/ethan-goldstein/Autonomous-OS

Remote

- Designed an 11-agent orchestration platform on a Node and Express ESM core where every agent inherits scheduling, SQLite persistence, and live log streaming from a shared base class, so adding an agent is one file and one registry line.
- Built a two-tier LLM dispatcher, headless Claude Code CLI primary with local Ollama fallback behind a 10-minute persisted circuit breaker, and capped inference at 2 concurrent processes on a FIFO semaphore after unbounded forking crashed an 8GB machine.
- Persisted the fleet on node:sqlite with no ORM across 13 tables, adding WAL journaling and a 5-second busy timeout after concurrent writes began failing boots, and streamed live state to a React dashboard over Server-Sent Events.
- Routed every outward-facing action through one egress queue where risky lanes require a human tap that no toggle can override, and funnelled all model calls through a single chokepoint that treats fetched content as data and never as instructions.
- Wrote HMAC-SHA256 session auth on node:crypto alone behind a prefix allowlist mounted before any route is declared, with no localhost bypass because Tailscale terminates TLS on loopback; deployed via launchd with no public port.
- Made automated outreach defensible: per-lane daily caps dripped across staggered windows, randomized send jitter, DNS MX pre-validation that fails open, and an IMAP bounce loop retiring dead addresses.

GovCIO

2026 to Present

Data Processor

Tysons Corner, VA

- Hold an active Public Trust clearance (U.S. Treasury, IRS) extended to Department of Veterans Affairs systems by reciprocity.
- Process confidential Veteran records under the VA HTMS contract in secured remote environments.

Oxford Government Consulting

2025

Data Entry Operator

Tysons Corner, VA

- Supported the IRS Digitalization-as-a-Service program with GovCIO, on a pipeline processing 2 million document images per day.

The Baseball Zone, Manager. Rebuilt the company website; ran front desk operations, scheduling, and daily finances. 2021 to 2024

PROJECTS

PARALLAX: Bitemporal Analytical Engine C++20, WebAssembly, SIMD, libFuzzer

- Query language written end to end, lexer through cost-based planner, over a columnar bitemporal store with a Z-order index. Policy refusal at plan time, sound because two of four cardinality estimators are exact. Differential tested against a naive oracle.

NULLHYP: Quantitative Research Engine TypeScript, Web Workers, zero-dependency ML, Python cross-check

- Reports no edge: Sharpe 2.46 in sample, -0.18 out, deflated Sharpe 0.06 over 12 trials. Validation came before any model: purged K-fold with embargo, combinatorial purged cross-validation, and a lookahead audit that recomputes every feature on data truncated at bar t. Trees, elastic net, and logistic regression from scratch in a Web Worker, matched on 22 of 22 checks by a pandas build.

WiFi Sensing Lab Python, signal processing, statistics, privacy engineering

- The 0.5 to 3 Hz gait band is unobservable at consumer polling rates, so motion detection from RSSI alone moved to the 0.1 to 0.8 Hz body-shadowing envelope, each link learning its own noise floor from a tracker that falls fast and rises slowly. Median AUC 0.676 over 10 links against a permutation null at 0.503, $p = 0.017$. The public repository contains no network client.

Real-Time Multiplayer Engine WebSockets, Cloudflare Durable Objects, Three.js, MediaPipe

- Authoritative game server on edge compute: fixed 60Hz simulation with seeded RNG decoupled from render. Three public titles.

Vector Stroke-Rendering Engine React, Vite, Three.js, GitHub Actions

- Auto-traces a raster logo into vector contours with marching squares, replayed against a Three.js pen. Live on a paying client's site.

SKILLS

Languages & Verification: C++20, Java, Python, TypeScript, SQL, Bash; libFuzzer, ASan and UBSan, differential testing

AI, ML & Statistics: multi-agent orchestration, local LLM inference (Ollama), Claude Code CLI, prompt-injection mitigation; gradient boosting, elastic net and logistic regression from scratch, purged and combinatorial cross-validation, permutation tests, DSP

Backend & Distributed Systems: Node.js, Express, WebSockets, Server-Sent Events, node:sqlite (WAL), REST, OAuth 2.0, HMAC-SHA256 auth, Cloudflare Workers and Durable Objects, Tailscale, launchd and cron, Playwright, Git, GitHub Actions CI/CD

Frontend: React, Next.js App Router, Vite, Three.js, WebGL, WebAssembly, GSAP, MediaPipe, code splitting, WCAG accessibility

Clearance: Public Trust (Active), U.S. Department of the Treasury (IRS); VA reciprocity granted 2026